# Logic and Reasoning Mastery Workbook

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# I have neither given nor received unauthorized assistance.

# 1 Problem 1

(20pts) Translate the tablets below and answer the questions on the next page. Submit just the second page with your solutions

- Describe your process of solving in detail.

  In order to solve this problem, we first needed to make some assumptions (or propositions as referred to in Rosen Section 1.1) about the tablet.

  1. We assume that the symbols represent numbers in a base-10 system.

  2. We assume the numbers displayed side by side is representative of a relationship between them (i.e. there is a relationship between each pair).

  3. The symbol on the first line represents the number 1. With these assumptions, we proceeded to attempt to decode the symbols into standard base-10 numbers. We represet each of the pairs in tuples, defined below:

$$\begin{aligned} T = \{&(1,9),(2,18),(3,27),(4,36),\\ &(5,45),(6,54),(7,63),(8,72),(9,81),\\ &(10,90),(11,99),(12,108),(13,117),(14,126)\} \end{aligned}$$

  From here, we noticed that the second number in each tuple is simply 9 times the first number. This led us to conclude that the relationship represented by the symbols is a multiplication by 9.

- Are you using inductive or deductive reasoning?

  Per the definitions provided in the workbook, inductive reasoning is the method of drawing general conclusions from a limited set of observations, while deductive reasoning is the method of using logic to draw conclusions from statements we accept as true. We are using inductive reasoning. We observed specific instances (the pairs of numbers) and identified a pattern (multiplication by 9) that we generalized to form a conclusion about the relationship represented by the symbols.

- Can you be sure your answer is correct?

  While we cannot be absolutely certain without additional context or confirmation, the consistent pattern observed across all pairs strongly suggests that our conclusion is correct. The simplicity and uniformity of the relationship (multiplication by 9) across all provided pairs lends credibility to our solution.

## 2 Problem 2

In the Rosen Book, page 8, buried in a bunch of words are some essential ideas we will use again and again this semester. You should know these and be ready. (10pts) Write the definitions of converse, contrapositive, and the inverse in a format that is useful for you to reference - make a poster, index cards, or even make notes in your book. Include a photo of this in your homework on this page (small is fine).

Sorry I needed the photo to be a separate page! I don't have the PDF editor to embed it into the document directly.

"If Juan has a smartphone, then $2 + 3 = 5$"

is true from the definition of a conditional statement, because
value of the hypothesis does not matter then.) The conditional

"If Juan has a smartphone, then $2 + 3 = 6$"

is true if Juan does not have a smartphone, even though 2 +
these last two conditional statements in natural language (exc
there is no relationship between the hypothesis and the conclu
ematical reasoning, we consider conditional statements of a r
English. The mathematical concept of a conditional statemen
effect relationship between hypothesis and conclusion. Our def
specifies its truth values; it is not based on English usage. Prop
language; we only parallel English usage to make it easy to us

The if-then construction used in many programming lang
in logic. Most programming languages contain statements su
proposition and $S$ is a program segment (one or more statements
of a program encounters such a statement, $S$ is executed if $p$ i
is false, as illustrated in Example 8.

In order to illustrate the converse, contrapositive, and inverse, we can use the following basic proof to illustrate it

If joe does his homework then joe will get a good grade

this is our p -> q, then

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converse: if joe gets a good grade then joe did his homework

contrapositive: joe will not get a good grade if joe does not do his homework

inverse: if joe will not do his homework then he will not get a good grade

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converse is proving a standard conditional statement backwards implys the standard conditional statement is true

contrapositive is proving the opposite of a standard conditional statement backwards implys the standard conditional statement is true

inverse is proving the opposite of a standard conditional statetment implys the standard conditional statement is true

**EXAMPLE 8** What is the value of the variable $x$ after the statement

$$\textbf{if } 2 + 2 = 4 \textbf{ then } x := x + 1$$

if $x = 0$ before this statement is encountered? (The symbol
statement $x := x + 1$ means the assignment of the value of $x$

*Solution:* Because $2 + 2 = 4$ is true, the assignment statemen
$x$ has the value $0 + 1 = 1$ after this statement is encountered.

**CONVERSE, CONTRAPOSITIVE, AND INVERSE** We
statements starting with a conditional statement $p \to q$. In p
conditional statements that occur so often that they have specia
is called the **converse** of $p \to q$. The **contrapositive** of $p \to$
The proposition $\neg p \to \neg q$ is called the **inverse** of $p \to q$.
conditional statements formed from $p \to q$, only the contrap
value as $p \to q$.

We first show that the contrapositive, $\neg q \to \neg p$, of a con
has the same truth value as $p \to q$. To see this, note that the
$\neg p$ is false and $\neg q$ is true, that is, only when $p$ is true and $q$ is false. We now show that neither the converse, $q \to p$, nor the inverse, $\neg p \to \neg q$, has the same truth value as $p \to q$ for all possible truth values of $p$ and $q$. Note that when $p$ is true and $q$ is false, the original conditional statement is false, but the converse and the inverse are both true.

Remember that the contrapositive, but neither the converse or inverse, of a conditional statement is equivalent to it.

When two compound propositions always have the same truth value we call them **equivalent**, so that a conditional statement and its contrapositive are equivalent. The converse and the inverse of a conditional statement are also equivalent, as the reader can verify, but neither is equivalent to the original conditional statement. (We will study equivalent propositions in Section 1.3.) Take note that one of the most common logical errors is to assume that the converse or the inverse of a conditional statement is equivalent to this conditional statement.

We illustrate the use of conditional statements in Example 9.

# 3 Problem 3

(10 pts) Write up your own conditional example, like "If it is raining, the grass is wet." but more interesting and memorable. Write the converse, contrapositive, and the inverse of your example.

- Original Statement: If a student studies hard, then they will pass the exam.
- Converse: If a student passes the exam, then they studied hard.
- Contrapositive: If a student does not pass the exam, then they did not study hard.
- Inverse: If a student does not study hard, then they will not pass the exam.

For my own conditional statement, I will choose something applicable to my life as a coffee enthusiast:

- Original Statement: If I drink coffee in the morning, then I feel more awake during the day.
- Converse: If I feel more awake during the day, then I drank coffee in the morning.
- Contrapositive: If I do not feel more awake during the day, then I did not drink coffee in the morning.
- Inverse: If I do not drink coffee in the morning, then I do not feel more awake during the day.

## 4 Problem 4

(10pts) Construct the Truth Tables from Rosen page 15 #31.

In Rosen page 15 #31, we must construct truth tables for the following statements:

a) $p \wedge \neg p$

| p | $\neg p$ | $p \wedge \neg p$ |
|---|---|---|
| T | F | F |
| F | T | F |

b) $p \vee \neg p$

| p | $\neg p$ | $p \vee \neg p$ |
|---|---|---|
| T | F | T |
| F | T | T |

c) $(p \vee \neg q) \rightarrow q$

| p | q | $\neg q$ | $p \vee \neg q$ | $(p \vee \neg q) \rightarrow q$ |
|---|---|---|---|---|
| T | T | F | T | T |
| T | F | T | T | F |
| F | T | F | F | T |
| F | F | T | T | F |

d) $(p \vee q) \rightarrow (p \wedge q)$

| p | q | $p \vee q$ | $p \wedge q$ | $(p \vee q) \rightarrow (p \wedge q)$ |
|---|---|---|---|---|
| T | T | T | T | T |
| T | F | T | F | F |
| F | T | T | F | F |
| F | F | F | F | T |

e) $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$

| p | q | $p \rightarrow q$ | $\neg q$ | $\neg p$ | $\neg q \rightarrow \neg p$ | $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$ |
|---|---|---|---|---|---|---|
| T | T | T | F | F | T | T |
| T | F | F | T | F | F | T |
| F | T | T | F | T | T | T |
| F | F | T | T | T | T | T |

f) $(p \rightarrow q) \rightarrow (q \rightarrow p)$

| p | q | $p \rightarrow q$ | $q \rightarrow p$ | $(p \rightarrow q) \rightarrow (q \rightarrow p)$ |
|---|---|---|---|---|
| T | T | T | T | T |
| T | F | F | T | T |
| F | T | T | F | F |
| F | F | T | T | T |

# 5 Problem 5

5. (15pts) Imagine you are given 3 matchboxes, labeled "Red and White," "Red and Red," and "White and White." Each box contains 2 marbles which can be either red or white. The labels on each box are incorrect (although the labels would be correct if you switched them around). You are allowed to peek inside one box and look at exactly one marble in that box and see if it is red or white.

1. After this, can you figure out what is in each box and fix the labels to be correct?

**Yes**, if we break down the problem, all labels are incorrect. This means that if you open the box labeled "Red and White", whatever marble you see you know is either "Red and Red" or "White and White". Then, once you know whether the box is "Red and Red" or "White and White", you can deduce the other boxes just by swapping the labels since we know the other box is also wrong. In order to illustrate this, lets say we selected a red marble from the box labeled "Red and White". This means that this box must be "Red and Red". Therefore, the box labeled "White and White" must actually be "Red and White", and the box labeled "Red and Red" must actually be "White and White".

2. Does it matter which box you choose to open at the start?

Yes, you should choose the box labeled "Red and White", because whatever marble you select from that box you know will give you enough information to classify the box correctly (see reasoning in question 1). If you were to select either of the other boxes first, you would not have enough information to deduce the contents of all boxes.

3. Are you using inductive or deductive reasoning?

Here we are using deductive reasoning. We are using logic to draw conclusions from the information we have (the incorrect labels and the color of the marble we see) to determine the correct contents of each box.

4. Can you be sure of your answer?

Yes, we can be sure of our answer because the reasoning is based on logical deduction from the initial conditions (all labels are incorrect) and the observation made (the color of the marble seen). The process leads to a definitive conclusion about the contents of each box.

# 6 Problem 6

(15pts) At a famous pizza place, Albert, Bob, and Jose are asked by a waiter, "Does everybody want water?" Albert says "I don't know." Bob then says, "I don't know." Jose says, "Not everyone wants water." The waiter returns and brings correctly each person what they wanted. Which people got water?

Let us use prepositional logic to break down this problem. Let A represent "Albert wants water", B represent "Bob wants water", and J represent "Jose wants water". The waiter's question can be represented as $A \wedge B \wedge J$ (Does everybody want water?). Based on Jose's response, we know that not everyone wants water, therefore, at least one of A, B, or J is false. We also know that there is only one configuration total that represents the interests of all three people. Now assuming Jose is not a mindreader, we can say that Jose does not want water. This is because if both Albert and Bob wanted water, Jose would have known that everyone wanted water and would not have said "Not everyone wants water." Therefore, we can conclude that Jose does not want water (J is false). Also, since Bob answered second with "I don't know", we can conclude that Bob wanted water. This is because if Bob did not want water, then he would have said "Not everyone wants water" just like Jose did. Therefore, we can conclude that Bob wants water (B is true). Finally, since Albert answered first with "I don't know", we can conclude that Albert also wanted water. This is because if Albert did not want water, then he would have said "Not everyone wants water" just like Jose did. Therefore, we can conclude that Albert wants water (A is true). In summary, Albert and Bob got water, while Jose did not.

# 7 Problem 7

(10 pts) Let's do a simple deductive proof that a negative number times a negative number is a positive number.

Prove (-a)(-b) = ab.

To prove this, we will use the same properties as displayed in the mastery workbook.

We may assume the following properties:

- Basic principles of algebra as defined in Appendix A and videos.
- FOIL
- A negative number times a positive number is negative.

Using these properties, we show that:

1. $0 = 0 \cdot 0$ (Property of Zeros)
2. $0 = (a - a) \cdot (b - b)$ (By the definition of inverses)
3. $0 = ab - ab - ab + (-a)(-b)$ (By FOIL and multiplication)
4. $ab = ab - ab + (-a)(-b)$ (Adding ab to both sides of the equation via basic algebra)
5. $ab = (-a)(-b)$ (By simplification via subtraction of the expression $ab - ab$ (using basic algebra))

Thus, we have shown that (-a)(-b) = ab. $\square$

# 8 Problem 8

(10 pts) Compare the proof in the previous question to the car example video in Moodle. Both show "a negative times a negative is a positive."

- Which method is inductive reasoning?

  The car example video uses inductive reasoning. It uses the car and its relative position as a real world analogy to illustrate the concept of multiplication of negative numbers using the relative position and direction of a car in Colorado, where Denver is the origin, and north and south are positive and negative directions respectively. By observing the car's movement in this context, we can generalize the concept that a negative times a negative results in a positive.

- Which method is deductive reasoning?

  The algebraic proof in the previous question uses deductive reasoning. It starts with known axioms and applies logical steps to derive the conclusion that (-a)(-b) = ab.

- Which method is more convincing to you?

  For me the algebraic proof is more convincing for the purpose of multiplication of negative numbers. This is because simple algebraic axioms can more easily draw you to the desired result without a real world analogy. However, I do think that for actually defining and understanding axioms themselves (like understanding the definition of 0 or multiplication), inductive reasoning is much more useful.